Qualifying Fusion Structural Materials Without the Irradiation Data

a method for the regime where the data cannot be obtained at all

A self-initiated track: the regime where extrapolation is documented to fail · every number carries a cloud job receipt
Public repository: https://github.com/sharadbachani-oss/merlin-quantum-fusion-materials

Executive summary

3.074 meV	is what it costs to delete **every measured constant** from the calculation. Same basis, same method, same geometry, constants derived from the model's own structure instead of from CODATA. That is **fourteen times inside chemical accuracy**, so the missing irradiation data was never load-bearing for the constants layer
–0.19%	on the bond length of the fuel molecule from zero empirical input. Hydrogen isotope retention sits directly in the DEMO safety case, and this is computed with nothing fitted and no measured constant entering
151% FAIL	on the helium repulsive wall, reported at the same receipt class as the passes. Second-order theory in a double-zeta basis is not adequate for it. That is a solver limit, not a constants limit, and it is named as the build task

Why this domain, and why now

First-wall materials must be qualified against a **14 MeV neutron flux no existing reactor can produce**. The field's own assessment is that extrapolation from fission data fails, because there are no observations in the operating range. IFMIF-DONES is being built to manufacture the missing spectrum and it sits **on the critical path of DEMO**. Its purpose is to generate the parameters damage models need. We are not proposing that it should not be built — irradiation validates, it is not replaced by a model. We are proposing that the **schedule dependency can be broken**: a parameter-free model can be frozen now, and the facility then tests it rather than supplying it.

A parameter-free model in a data-rich domain buys validation cost. In a domain where the data cannot be obtained it is the difference between having a method and having none. Fitted models and machine learning win where data is abundant; this wins where data is impossible. Every number here is reproducible in about a minute with no credentials (`python verify.py`, 7 of 7), and the one failure ships alongside the successes.

Technical Report

The problem, in the field's own words

First-wall and structural materials in a fusion power plant see a **14 MeV neutron flux that cannot be created in any existing test reactor**. Fission reactors and spallation sources produce a different damage spectrum, and the assessment is blunt: attempts at extrapolation are unsuccessful, because there are no experimental observations in the operating range of a fusion reactor.

The response has been to build the missing experiment. **IFMIF-DONES** is a dedicated accelerator driving a deuteron beam into a liquid-lithium curtain to manufacture a fusion-like neutron spectrum, and it is described in the European programme as sitting **on the critical path of DEMO**. Design, licensing and safe operation are gated on it.

That facility exists to produce numbers. Displacement thresholds, helium and hydrogen binding and migration energies, cluster and void energetics. Every damage model needs them, and every damage model currently gets them by measurement or by fitting to measurements that do not exist at fusion conditions.

This track asks a different question: what if the model needs no measured constant at all?

Why this is structurally different: nothing is fitted

Our engine's operator constants follow from the model's own structure and are fixed before any material or irradiation data is seen. In a data-rich domain that buys validation cost. **In a domain where the data cannot be obtained, it is the difference between having a method and having none.**

That is the whole thesis of this track, and it reduces to one testable question. If you delete every empirical constant from a calculation, does the answer change? If it does, the approach is dead. If it does not, the missing irradiation data was never load-bearing for the constants.

Gate 1 — deleting every measured constant costs 3 meV

We run the identical molecular calculation twice: once on measured (CODATA) constants, once on constants derived from the model's own structure. Same basis, same method, same geometry. The only difference is where the constants came from, and we publish only that difference: the two absolute energies side by side would let a reader back out the ratio of the constant sets, and the difference alone makes the identical point.

separation	difference, derived vs measured constants	vs chemical accuracy
0.7414 Å	**0.099 meV**	0.002×
1.0000 Å	2.201 meV	0.051×
1.4000 Å	**3.074 meV**	0.071×

Worst case across the curve is **3.074 meV, which is fourteen times inside chemical accuracy** (1 kcal/mol = 43.4 meV).

Removing the entire empirical input layer changes nothing that a materials model can resolve. The constants were never the reason the data was needed.

Gate 2 — the fuel molecule, from zero empirical input

Hydrogen is not an academic choice here. It is the fuel, and hydrogen isotope retention in the first wall is a licensing-critical quantity, because tritium inventory limits sit directly in the safety case.

quantity	derived-constant result	experiment	deviation
bond length R_e	0.7400 Å	0.7414 Å	–**0.19%**
dissociation energy D_e	4.386 eV	4.478 eV	–2.1%

Nothing was fitted, and no measured constant entered. The residual is the correlation level, not the constants, which Gate 1 has already bounded at 3 meV.

Gate 3 — the boundary, reported rather than smoothed

We also tested the quantity that governs **helium bubble pressure**, the short-range helium repulsive wall. Helium is the fusion-specific damage species: a 14 MeV spectrum produces roughly a hundred times more helium per displacement than a fission spectrum, and helium embrittlement is the failure mode that fission experience does not cover.

separation	derived-constant result	accurate ab initio	ratio
1.00 Å	4.130 eV	1.429 eV	2.89
1.25 Å	1.491 eV	0.571 eV	2.61
1.50 Å	0.521 eV	0.256 eV	2.04

Mean deviation **151%**. At second-order perturbation theory in a double-zeta basis this calculation is **not adequate** for the helium wall, and we will not present it as if it were.

This is a correlation-level and basis-set limit, not a constants limit. Gate 1 shows the constants agree to 3 meV, so the gap is in the solver. It is a defined build task with a known answer in the literature: coupled cluster with a triple-zeta or explicitly-correlated basis. We name it here because a track that only reports its successes is not worth reviewing.

What this is worth, and to whom

The comparison is not against a better simulation. It is against a facility.

route	what it costs	what it delivers
IFMIF-DONES	dedicated accelerator and lithium loop, on DEMO's critical path	a fusion-like neutron spectrum, and the parameters measured from it
derived-constant modelling	compute	the same parameters, with no irradiation step

We are not proposing that the facility should not be built. Irradiation experiments validate; they are not replaced by a model. What we are proposing is that **the schedule dependency can be broken**: a parameter-free damage model can be built and frozen now, and the facility then tests it rather than supplying it.

For a programme where materials qualification is the critical path, moving a model from *downstream of the data* to *independent of the data* changes the sequencing of the whole plant.

What we do not claim

- **No irradiation prediction yet.** Gates 1 and 2 establish that the constants layer is sound and that the empirical input can be deleted. They do not yet compute a displacement threshold or a helium binding energy in a metal lattice.
- **Not a solid-state engine today.** The current implementation is molecular. Extending to periodic systems is the second build task.
- **Gate 3 is a fail.** Reported as one.
- **No claim that experiment is unnecessary.** The claim is that the model need not wait for it.

The blind prediction, frozen

Every other number in this document is a postdiction against a known value. This one is not.

Species. Be■H■, the first oligomer between monomeric beryllium hydride and the polymeric solid. Beryllium is the ITER first-wall material and neutron multiplier, and beryllium-hydride chemistry governs tritium retention in it. The monomer BeH■ has been measured to **r_e = 1.326407(3) Å** in a discharge-furnace source. For the dimer we could locate **no experimental gas-phase structure determination**, and the same apparatus that produced the monomer is a plausible route to it. That makes this a prediction somebody can go and check.

Frozen 2026-09-09T15:02:11Z, SHA-256 `cc9092bfa29e5e522c3a5cab57b048cc0f8f87378fefd4f7da0a3623d7bba497`**.**

quantity	prediction	band
r(Be–Be)	2.051 Å	±0.08
r(Be–H) bridging	1.505 Å	±0.06
r(Be–H) terminal	1.326 Å	±0.02

quantity	prediction	band
angle Be–H_b–Be	85.9°	±4.0
dimerisation energy, 2 BeH■ → Be■H■	–1.04 eV	±0.50

How the bands were set, and why they differ. The method is short by 1.99% on the measured monomer bond, so the raw geometry is scaled by that calibration and the uncertainty reflects how far the transfer can be trusted. Tightest on the terminal bond, which is the calibrant's own bond type. Loosest on the three-centre bridge and the Be–Be separation, which are not.

Falsified if the equilibrium structure is not the D2h bridged form, or any distance falls outside its band, or the dimerisation energy has the opposite sign.

Scoring rule, stated in advance. Score every entry, not a subset. A hit on the terminal bond alone is not a success, because that bond is the easy case and shares its type with the calibrant.

What is and is not blind here. Computational values for this species exist in the literature at DFT and composite-method level. The registration is blind against **experiment**, not against theory, and it says so in the frozen record. If an experimental determination already exists and we did not find it, this is a postdiction and should be scored as one.

Why a parameter-free method can do this at all. Nothing in the calculation was fitted to a measured range, so there is no range outside which it falls silent. A model fitted to existing data cannot make this move, and in fusion materials the operating range contains no data at all. That is the entire argument of this track, reduced to five numbers and a hash.

Prediction v2 — better basis, same discipline

v1 was computed in a minimal tabulated basis, which the diborane validation showed was the limiting factor. v2 changes **only the basis**, and the choice is deliberate and not the obvious one.

Why not 6-31G\\ or a similar standard basis. Tabulated basis sets carry exponents optimised against experimental data. Using one would inject fitted parameters into a track whose entire claim is that nothing is fitted. v2 uses the engine's **first-principles generated basis** instead, which is constructed rather than fitted — and is also simply better: on H■ it gives –0.19% against –1.54% for the tabulated alternative.

The calibrants improve sharply.

calibrant	v1 offset	v2 offset
BeH■	–1.99%	**+0.27%**
BH■	–0.84%	+0.84%

And so does the validation. Rerunning the diborane test at the v2 level:

quantity	v1 error	v2 error
r(B–B)	+0.0654 Å	–**0.0148 Å**
r(B–H) bridging	+0.0147 Å	–0.0245 Å
r(B–H) terminal	–0.0095 Å	–**0.0043 Å**

Worst error falls from 0.0654 Å to 0.0245 Å. The v2 bands are set at roughly twice the error this level actually made on the measured analogue, which is why they are tighter than v1's rather than tighter by assertion.

v2, frozen 2026-09-09T22:13:39Z, SHA-256 `e05a671dc31387970b0819be539752552cbc024712e29454223fcd4d776b54f6`**.**

quantity	v1	v2	band
r(Be–Be)	2.051 Å	**2.005 Å**	±0.03
r(Be–H) bridging	1.505 Å	**1.460 Å**	±0.05
r(Be–H) terminal	1.326 Å	**1.321 Å**	±0.01
angle Be–H_b–Be	85.9°	**86.7°**	±3.0
dimerisation energy	–1.04 eV	–**1.15 eV**	±0.50

The energy band is deliberately **not** tightened: no measured dimerisation energy was used to validate this level, so tightening it would be assertion.

The result worth noticing. v2 moved the Be–Be distance down by 0.046 Å, which is the direction the diborane bias predicted — and **no bias correction was applied**. The shift came from the basis alone. Two independent routes, an analogue's measured error and an improved calculation, point the same way. Every v2 value also sits inside v1's bands, so the two levels agree.

v1 is not withdrawn. It stands frozen exactly as registered, and the two are scored separately. Reporting only whichever lands closer would be the same error as bias-correcting, arriving later. That instruction is written into both frozen records.

Method validation on known data

The prediction above rests on one assumption: that a systematic offset measured on a monomer transfers to its dimer. That assumption is testable on a case where **both** are measured, so we tested it before claiming anything.

The analogue. Diborane. B■H■ is the direct structural counterpart of Be■H■, the same electron-deficient bridged hydride, and both it and its BH■ monomer are experimentally characterised. We ran the **identical** technique: same theory, same basis, same derived constants, same calibrate-on-the-monomer-and-transfer procedure, same uncertainty bands, and a neutral starting geometry rather than one seeded from the answer.

quantity	predicted	measured	error	band	inside
r(B–B)	1.835 Å	1.770 Å	+0.065	±0.08	yes
r(B–H) bridging	1.349 Å	1.334 Å	+0.015	±0.06	yes
r(B–H) terminal	1.178 Å	1.187 Å	–0.010	±0.02	yes

Three of three inside the bands, using the same bands quoted for the Be■H■ prediction. The technique does what it claims, and the bands are not decorative.

A second transfer test, on a different bond type. One analogue is a sample of one, so we ran the technique again on CH■ → C■H■, where both are measured to high precision and the carbon–carbon separation probes the same heavy-atom quantity that was weakest in diborane.

quantity	predicted	measured	error	band	band used
r(C–C)	1.537 Å	1.535 Å	+0.002	±0.08	**2%**
r(C–H)	1.087 Å	1.094 Å	–0.007	±0.02	35%

Two of two inside band, and the heavy-atom distance is near-exact.

Which sharpens the conclusion. The heavy-atom weakness is **not** a general property of the technique. On an ordinary two-centre bond the carbon–carbon separation lands at 2% of its band; on the three-centre bridged boron system it consumed 82%. The difficulty is specific to electron-deficient bridged bonding. Be■H■ is exactly that kind of system, so diborane is the governing analogue and the Be–Be entry remains the one most likely to sit near its edge.

Two things the validation exposes, and both are worth stating.

The calibration offset is **not a constant, and not even of fixed sign**: –1.99% on BeH■, –0.84% on BH■, **+1.20%** on CH■. It therefore has to be measured per system rather than assumed, and a single global scale factor would have been the wrong construction.

The bridged metal–metal distance is the weakest link, consuming **82% of its band** against 25% for the bridge bond and 48% for the terminal bond, while the unbridged carbon–carbon case used 2%. A scorer should read the Be–Be entry as the one most likely to sit near its edge.

One process note, because it nearly went the other way. The first ethane run returned a carbon–carbon distance of 1.9 Å and a flat failure on both entries. That was a sign error in our own geometry construction, which put the hydrogens on the wrong side of each carbon; a 1.9 Å C–C bond is not physical and the result was discarded and rerun rather than reported. The numbers above are from the corrected geometry.

The frozen prediction is unchanged. This validation was run after the freeze and adds evidence about the method, not a revision of the numbers. The hash in the record still covers exactly what was committed on 2026-09-09.

Receipt: `results/method_validation_b2h6.json`.

Reproduce it

``bash python verify.py # standalone, replays all 13 checks, stdlib only ``

`verify.py` has **no dependencies beyond the Python standard library** and no credentials. It replays every headline number in this document from the archived receipts and re-derives the blind prediction's SHA-256 from its own payload, so the freeze can be checked by anyone.

``bash MERLIN_QC_ENGINE=/path/to/engine python fusion_constants_gate.py MERLIN_QC_ENGINE=/path/to/engine python freeze_prediction.py ``

These two regenerate the numbers from scratch and **require our in-house quantum-chemistry engine, which is not part of this repository**. That is stated plainly rather than implied: an external reader can verify every archived number and the frozen hash, and cannot re-run the solver. The prediction is falsifiable by experiment regardless, which is the point of freezing it.

Receipts: `results/fusion_constants_gate.json`, `results/PREREG_be2h4_blind.json`.